

RESPONSES OF BASS TO NATURAL AND ARTIFICIAL
TEMPERATURE REGIMES¹

Charles C. Coutant, P.D.
Environmental Sciences Division
Oak Ridge National Laboratory

For presentation and possible publication at the National Symposium
on the Biology and Management of Centrarchid Basses, Tulsa, Oklahoma,
3-6 February 1975.

MASTER

¹Research supported by the U. S. Atomic Energy Commission under contract
with the Union Carbide Corporation.

NOTICE

This report was prepared as an account of work
sponsored by the United States Government. Neither
the United States nor the United States Atomic Energy
Commission, nor any of their employees, nor any of
their contractors, subcontractors, or their employees,
makes any warranty, express or implied, or assumes any
legal liability or responsibility for the accuracy, com-
pleteness or usefulness of any information, apparatus,
product or process disclosed, or represents that its use
would not infringe privately owned rights.

Responses of Bass to Natural and Artificial
Temperature Regimes

Charles C. Coutant

Dr. Charles C. Coutant, is Project Supervisor for power plant effects studies in the Environmental Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN. He and his associates are actively studying thermal effects on behavior and growth of several fish species and invertebrates. He publishes annual summaries of thermal effects literature and has contributed to development of thermal criteria and power plant environmental impact assessments to protect aquatic life.

Introduction

Temperature is undoubtedly the environmental factor which most consistently influences the lives and habits of the freshwater centrarchid basses. As cold-blooded organisms they cannot escape it - it controls the reproductive cycle, it determines the rates of feeding and metabolism, it determines swimming performance and growth rate, it influences distribution in lakes and rivers. To the sportsman, all of these add up to three important characteristics: where the fish are, how well they're biting, and what condition they're in.

Over the entire life history of a fish, the responses to temperature vary as the eggs are laid and develop, as the fry grow to adults and as populations change. Thus, no single temperature is good or bad in itself, but must be viewed in the context of when it occurs in the annual cycle of seasons and in its timing with the fish's various life stages and physiological activities.

"Temperature" has also become a pollutant in recent years. Electricity generating stations (except hydroelectric) must release heat to the environment as part of the steam cycle that turns the generators. Power plant steam condensers generally have used surface waters for cooling, thus raising water temperature. Many people are concerned that such increases could be detrimental to aquatic life and to popular recreational fisheries. As a result, the electric utility industry is being required to install closed cycle cooling - at a penalty in cost and energy efficiency.

This paper summarizes the responses of the centrarchid basses to temperature, and emphasizes the temperature requirements for successful

populations. Most information is available on largemouth (Micropterus salmoides) and smallmouth (M. dolomieu) bass. Some is also available for the spotted bass (M. punctulatus). The other basses - redeye (M. coosae), Suwanee (M. notius), and Guadalupe (M. treculi) - have been largely ignored in the environmental literature and receive only passing mention here.

Spawning

We may begin our excursion through the thermal experiences in the life of a bass with the point where it all begins - the successful spawn.

Largemouth

Largemouth bass will begin spawning activity in the spring whenever the water temperature rises to about 60°F (15.5°C). Courtship behavior preceeds actual egg laying, so the exact spawning temperature will depend upon the duration of courtship, time for "building" a nest, and the rates of temperature change while these activities are underway. Vagaries of weather may start and stop the spawning sequence as temperatures are warmed and cooled. Observations in the wild and at fish hatcheries indicate that temperature is a key factor (particularly days of rising temperatures above 60°F), but that precise definition of a "spawning temperature" isn't realistic. Most studies cite 66-68°F (18.9-20°C), however. The species has been reported to be self-sustaining in Puerto Rican reservoirs where water temperatures do not cool to these levels, but average about 78°F (25.5°C) throughout the year (Robbins and MacCrimmon 1974).

Recent successes at spawning largemouth in laboratory tanks by A. R. Carlson and J. G. Hale (1972, 1973) at the EPA National Water Quality Laboratory in Duluth, Minnesota have shed additional light on spawning temperature requirements. Courtship behavior began in spring within 1-2 days after the water reached 64.4°F (18°C).¹ Spawning occurred at temperatures from 68.9 to 72.5° (20.5-22.5°C). Bass transferred from seasonal temperatures of 46.5 to 50°F (2.5-10°C) in December, January, February, May and June to 68°F (20°C) spawned successfully after varying times (6-39 days) presumably reflecting time needed for further development of ovaries.

Smallmouth

According to early studies in Ontario by A. L. Tester (1930), the spawning temperature of the smallmouth bass is near 62°F (16.7°C) or about 4 degrees (F) less than for largemouth. Later studies in the same region by G. E. Turner and H. R. MacCrimmon (1970) showed nest building and spawning in early June at 59-64.4°F (15-18°C). Other studies across the range of the smallmouth have largely confirmed these temperatures and short nest-building periods. Spotted bass were intermediate in hatchery experiments by J. W. Howland (1932).

¹Most studies were conducted using the Celsius (Centigrade) temperature scale. Fractions of Fahrenheit degrees quoted therefore, do not usually represent a high level of accuracy, rather conversion from round Celsius numbers.

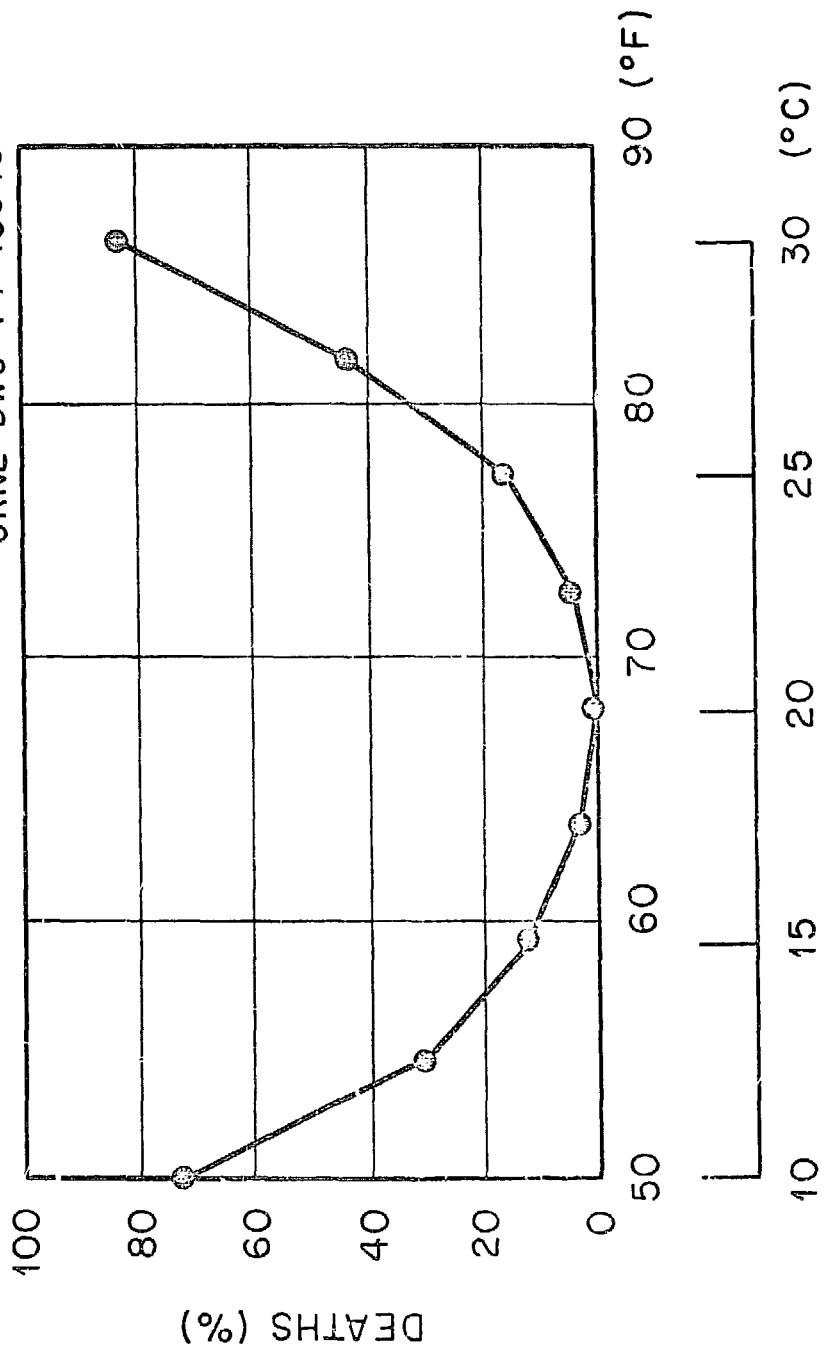
Incubation

Temperature greatly affects the development of bass eggs, by influencing both the rate of embryonic growth and the percentage survival. The incubation period for largemouth ranges from 48-72 hr, depending upon water temperature; for smallmouth it is about 4 days in the general spawning temperature range. The influence of temperature on survival appears to be the most important effect for determining the prominence of that year's offspring in the catches of subsequent years. The only environmental factor of greater importance seems to be the catastrophic effects of high winds that physically disrupt nests.

Sudden or prolonged chilling of eggs during the spawning season, instead of the usual steady or gradually rising temperatures, has traditionally been linked by fishery workers to reduced hatches and poor production. Careful observations of largemouth nests in Minnesota by R. H. Kramer (Kramer and Smith 1960) showed poorer survival of eggs during periods of cooler temperatures after spawning, and complete dieoff below 50°F (10°C). Information on the exact causes of such losses was sought by K. C. Jurgens and W. H. Brown (1954) of the Texas Game and Fish Commission in the early 1950's, and later by A. W. Eipper of Cornell University and graduate students J. W. Kelley (1968), W. H. Swallow (1968) and T. R. Badenhuizen (1969). They found through laboratory experiments that incubation temperature affects survival in a parabolic way, with best survival near 68°F (20°C) and progressively poorer survival at the extremes (Fig. 1). Detailed work by Badenhuizen showed that losses at the extremes were greater when eggs and embryos received abrupt thermal shocks than when they were exposed to slow changes on the order of 1.8°F

Fig. 1. Relationship between incubation temperature and average depth of largemouth bass embryos to 3 days after hatching. Embryos were slowly warmed or cooled from a spawning temperature of 68°F (20°C). Deaths plotted were those calculated to be attributable to the experimental temperature treatments (adapted from Badenhuizen 1969).

ORNL-DWG 74-13045



(1°C) per hour. The period within the first day after fertilization was the most sensitive.

However convincing the direct temperature effects on embryo survival may appear, they do not explain extremely high mortality in nature after very small amounts of temperature change, particularly rapid cooling during spring cold fronts. Jurgens and Brown, and later Kelley, may have correctly concluded negligence on the part of the male bass. Low water temperature often results in desertion of the nest by the male, leaving eggs unfanned and more susceptible to predation, siltation (which also may accompany cool, windy periods in spring), suffocation, and spread of fungus. Kelley observed similar rapid egg losses when he experimentally excluded males from some nests during an otherwise excellent hatching season for other nests. The Texas studies recommended artificial heating of bass ponds such as has been practiced at their A. E. Wood State Fish Hatchery near San Marcos, as a way to insure high yields. It seems, then, that further research effort needs to be given to the response of male bass to dropping temperature in spring rather than to the direct effects of temperature on developing eggs themselves.

A similar pattern of egg survival (Webster 1948) and male desertion of nests (Henderson and Foster 1956) has been seen for smallmouth bass. Temperatures of 55 to 79°F (12.8-26.1°C) seem favorable for egg development, but desertion of the nest after chilling reduces survival.

Feeding and Digestion Rates

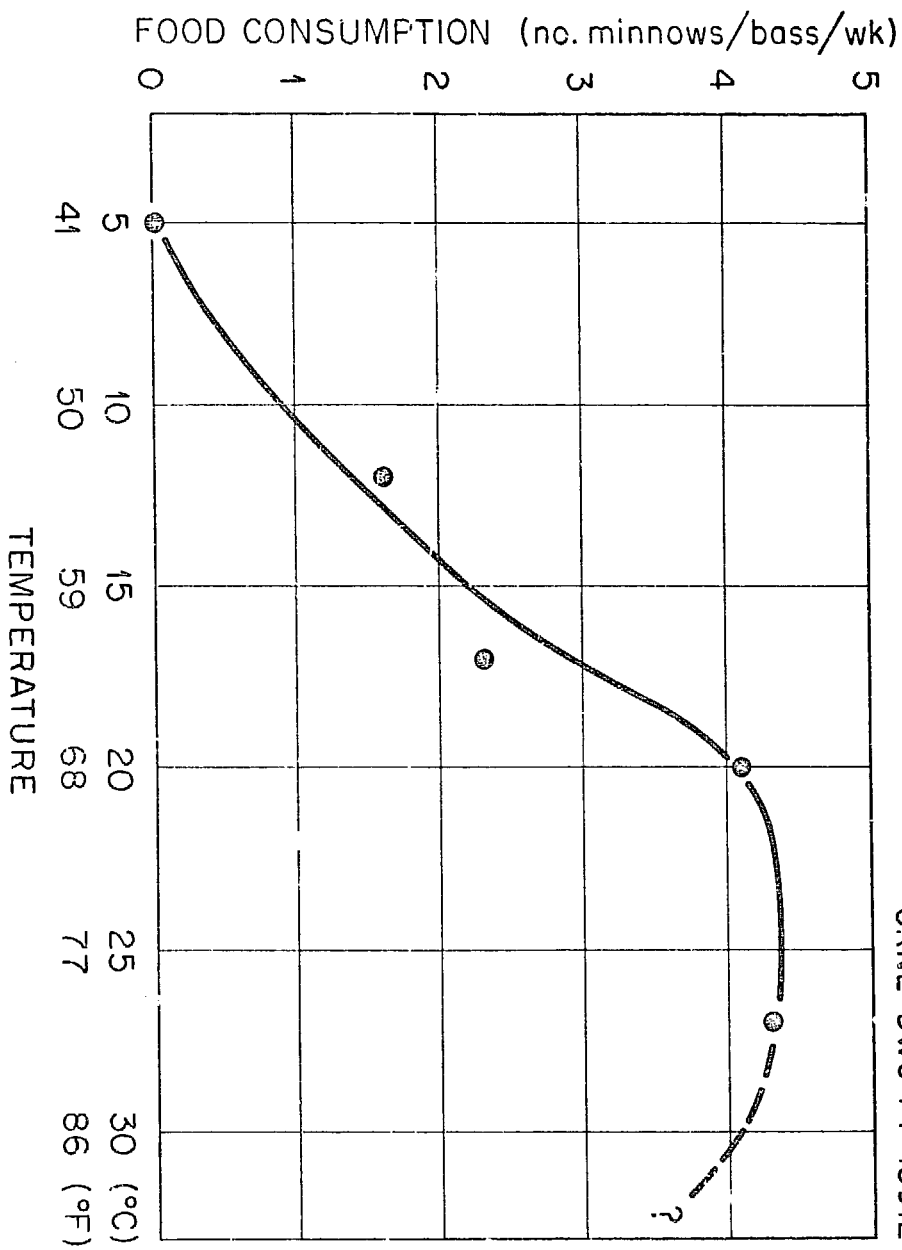
Feeding rate is greatly influenced by temperature. Below about 50°F (10°C) there is little, if any, natural feeding activity by either largemouth (Marcus 1932, Swingle 1949, Johnson and Charlton 1960, Keast 1968) or smallmouth (Webster 1954, Bennett and Childers 1957) bass. Between 50°F and near 68°F (10-20°C), the feeding rate increases rapidly with temperature rise (Fig. 2). Feeding occurs at a uniformly high rate to at least 80.6°F (27°C). Above that, feeding seems to decline, as seen both in the laboratory (Niimi 1973) and in the field, where adult bass in a Florida study remained quiet in shaded areas with their stomachs empty (Denyes and Joseph 1956). The decline at high temperature needs further checking, however.

Small fish may begin feeding at lower temperatures than larger fish as shown by the time annual rings are formed on scales (Stroud 1948). Marcus (1932) also noted in his experiments that the young fish fed more willingly at lower temperatures (50-65°F; 10-18.3°C) than did larger fish.

Detailed studies of digestion rates by largemouth at various temperatures by Molnar and Tolg in Hungary (1962) also show increasing activity with temperature rise. Their fish showed a logarithmic relationship with temperature: digestion at 77°F (25°C) occurred 5 to 6 times as fast as it did at 41°F (5°C). The fish had to be force fed at lower temperatures.

Fig. 2. Feeding rates of largemouth bass at different temperatures
(data after Johnson and Charlton 1960. The projected downward
trend is based on a few data by Niimi 1973.)

ORNL-DWG 74-13312



Growth

Many factors may affect the growth of fish, and temperature is only one. Others include quantity and quality of food, space limitations, the light cycle, diseases, parasites and pollution. These factors may interact, altering each other's effects, and their effects may not be the same on fish of different ages. Nonetheless, the effects of temperature on growth are often marked. When such fundamental necessities as food are available, temperature may be the principal factor determining growth rates and total annual growth.

Largemouth

Growth seems to begin in most largemouth populations near 50°F or 10°C (e.g., Bennett 1954, Marcus 1932). Young bass, according to Marcus, are able to begin growing at lower temperatures than adults. Further laboratory experiments could clarify this apparent age difference.

Growth rates of field populations of first-year largemouth bass in Minnesota were clearly shown by R. H. Kramer and L. L. Smith, Jr. (1960) to be faster at higher average daily water temperatures. The relationship was pronounced during the first 4 weeks of life, and when the total season's growth was compared, but it did not hold during the warmest part of the summer (after August 1st) probably because water temperatures rose above the growth optimum. Observations in warm reservoirs which provide cooling for electric power stations and other heat sources show higher than normal growth rates of young bass (Smith 1972, Graham et al. 1971, Bennett and Gibbons 1974, Busacker 1971). Rapid growth occurred in one hot reservoir arm (Bennett and Gibbons 1974)

even when recorded surface temperatures were very high (up to 101.5°F or 38.6°C in July). This would seem to have been due to young bass seeking more optimal temperatures in lower strata. Slowing of growth rates in wild fish at high summer temperatures has been observed, and correlated with accessory "checks" on fish scales used for determining fish ages (Beckman 1943, Bennett et al. 1940).

Detailed experimentation on temperature effects on growth has lagged considerably behind the suggestive field observations. K. Strawn (1961) followed the growth of largemouth fry at a number of temperatures. I replotted his data and compared them with those obtained by D. K. Cox and myself at Oak Ridge National Laboratory on fish near yearling size (Fig. 3). Growth rate increased in both cases to a maximum near 80.6°F (27°C). Above this "optimum" temperature, growth rate declined rapidly to zero near 95.9°F (35.5°C). Bass could continue to survive under the laboratory conditions for periods greater than one week only until the temperature reached 98.2°F (36.8°C). That is the "lethal temperature" as determined by J. C. Hart and others during the 1950's at the University of Toronto.

Fishery workers long ago discovered (see, for example, Bennett 1962) that southern latitudes, with generally warmer waters and longer growing seasons produced faster growth rates in warm-water species and often larger fish, particularly in largemouth bass (Fig. 4). South Florida largemouths reach such a large size that they have been considered as a separate subspecies. Analyses by J. P. Clugston (1964), then with the Florida Game and Freshwater Fish Commission, however, showed that the growth and size differences between northern and southern

Fig. 3. Growth rates of largemouth bass at various temperatures.
Upper: growth of fry plotted from data of Strawn 1961;
lower: growth of subadults of two size classes. C. C.
Coutant and D. K. Cox, Oak Ridge National Laboratory,
unpublished.

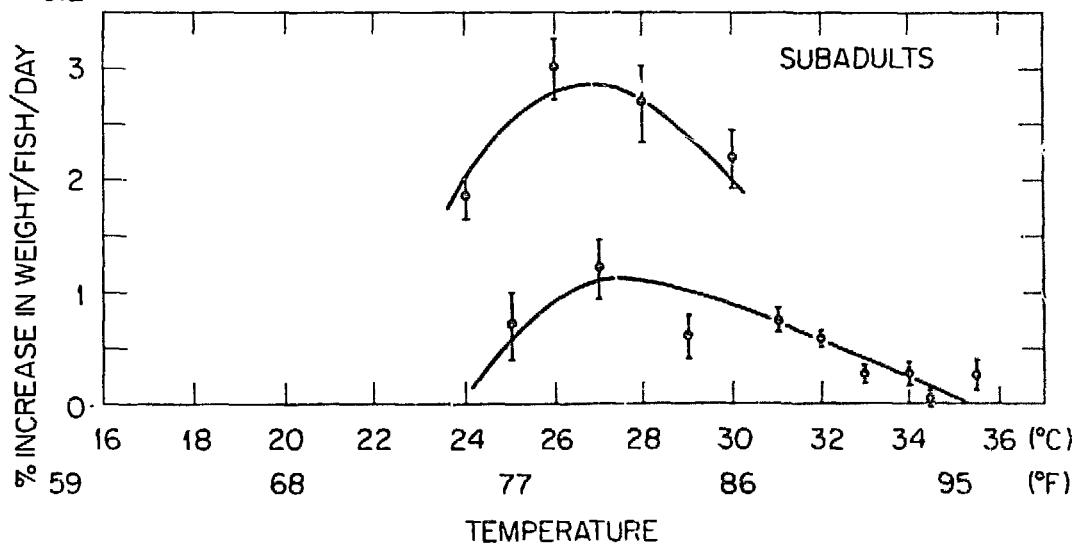
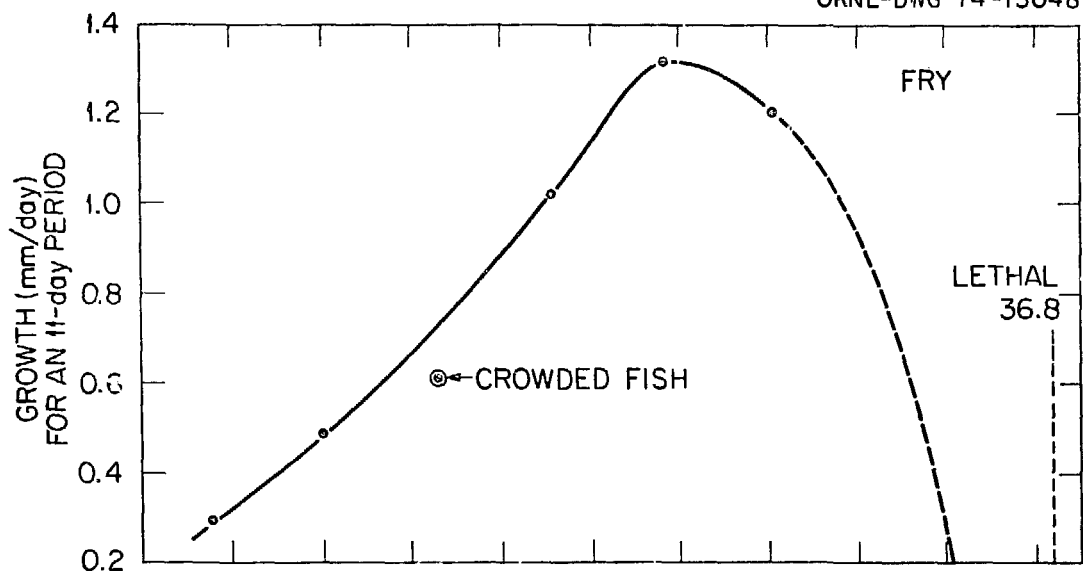
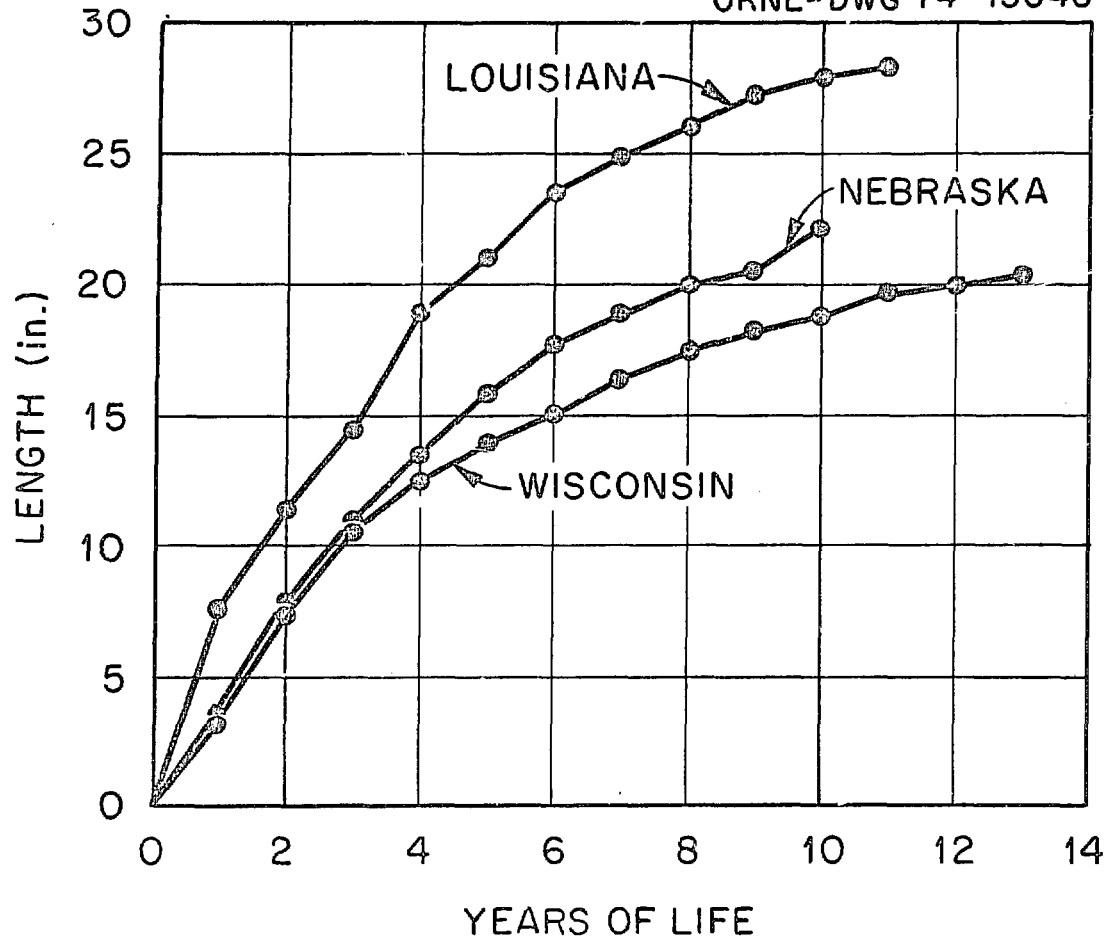


Fig. 4. Influence of latitude on growth of largemouth bass. Fish are compared from Louisiana (30 fish), Nebraska (25) and Wisconsin (618). Adapted from Bennet 1937.

ORNL-DWG 74-13046



largemouth were probably due to temperatures in the South suitable for growth year-round, and perhaps to poorer catchability (giving longer survival) of Florida fish rather than inherent genetic differences in growth potential. In fact, the body characteristics which were used to distinguish the Florida subspecies may have been due to temperature influences on embryonic development, a phenomenon common in many other fishes.

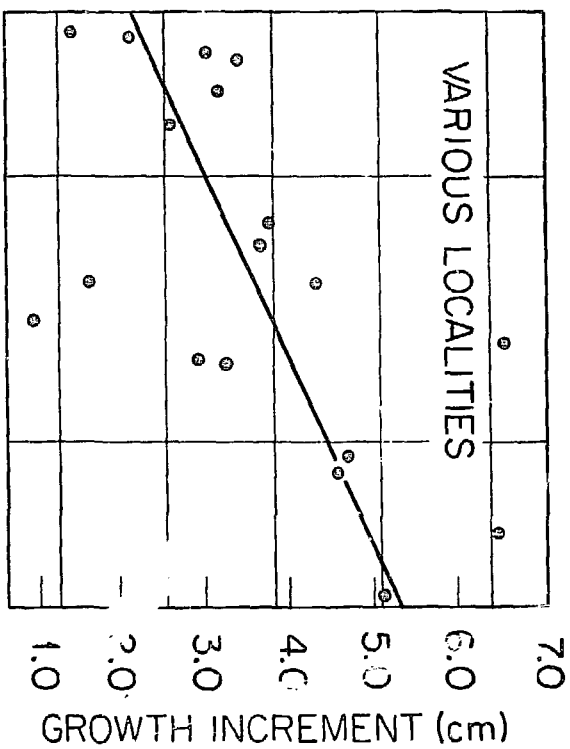
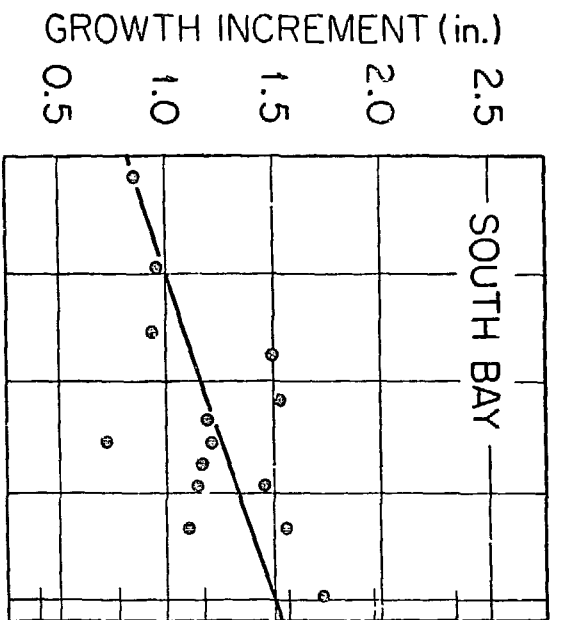
As Stroud (1948) pointed out, rapid growth rates of largemouth bass in the South seem to result in shorter lifetimes. Fish he studied in Norris Reservoir, Tennessee rarely exceeded 7 yrs. old while he noted 12 to 14 year-old fish in Connecticut and 15 year-old fish in Wisconsin and Minnesota. Another Tennessee lake yielded 11 year-olds as the oldest fish. Different catch rates (as discussed by Clugston 1964) may also affect longevity.

Smallmouth

D. W. Coble, then a graduate student at the University of Toronto, compiled data on annual growth for smallmouth bass in 19 populations throughout North America and compared growth with the average surface water temperature in each of the water bodies during July through September (1967). His results (Fig. 5) show a clear trend toward increased annual growth with higher summer temperatures. The large variability is undoubtedly due to influences of other factors. Stroud (1948) associated faster growth rates of smallmouth and spotted bass in southern reservoirs with earlier "senility" and earlier death from "old age" compared with cooler regions, as he had for largemouth.

Fig. 5. Average annual length increment of smallmouth bass of ages 3-5 in South Bay, Lake Huron, and in 18 populations throughout N. America in relation to mean surface water temperature for the period July through September. The slopes of the regression lines, which are placed by the method of least squares, are significantly different from zero (F-test; 95% level)(from Coble 1967).

ORNL-DWG 74-13049



16.5 17.0 17.5 18.0 18.5 17 20 25 28 (°C)
 61.7 62.6 63.5 64.4 65.3 62.6 68 77 82.4 (°F)

AVERAGE TEMPERATURE

GROWTH INCREMENT (in.)

GROWTH INCREMENT (cm)

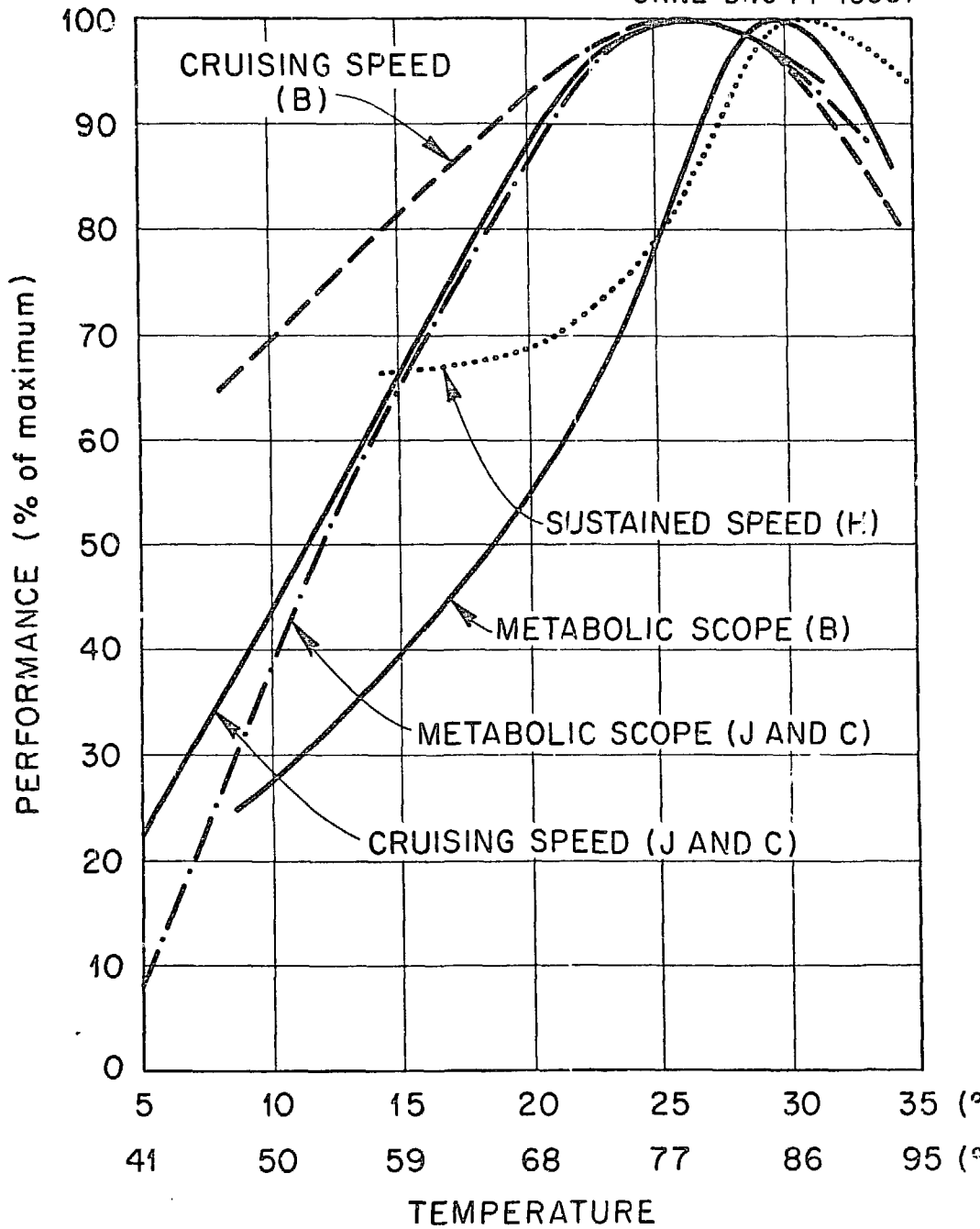
Experimental growth studies on smallmouth fingerlings (Peak 1965) and juveniles (Horning and Pearson 1973) show a temperature relationship similar in shape to that illustrated in Fig. 3 for largemouth. The growth optimum appears to be near 78.8°F (26°C) with a broad peak in the range 78.8-84.2°F (26-29°C). Growth is prevented above about 95°F (35°C). Both optimum and zero growth temperatures are only slightly less than those seen for largemouth.

Metabolism and Swimming Speeds

Water temperature influences the rates of most internal body functions of the basses as it does of all cold-blooded organisms. This is particularly true of the rate of oxygen consumption because it represents the summation of all the other processes. Largemouth bass fingerlings studied by Johnson and Charlton (1960) increased their rate of oxygen use 10 times between 41 and 86°F (5 and 30°C). Actively swimming fish used proportionately greater amounts of oxygen at all temperatures in that range. Beamish (1970) showed a similar trend for large bass, although the actual oxygen increase was not as great. To accommodate increases in oxygen demand by the fish's tissues, opercular beating increases. Denyes and Joseph (1956) observed that a 5-fold increase in opercular activity of largemouth was needed to keep the blood saturated with oxygen when temperatures were 89.6 to 95°F (32-35°C) instead of 41 to 50°F (5-10°C).

A particularly important feature of oxygen requirements in fishes has been called the "scope for activity." It is the difference between the oxygen needed by a resting fish at various temperatures and that

Fig. 6. Metabolic scope for activity and maximum swimming speeds of largemouth bass as a function of temperature determined by several investigators (J&C = Johnson and Charlton 1960; B = Beamish 1970; H = Hocutt 1973). Beamish used adults; the others used fingerlings.



needed during maximum exercise (swimming). As temperature rises, this "scope" or capacity available for exercise increases to a peak, then declines rapidly at stressing high temperatures (Fig. 6). Although experiments to date have not zeroed in on temperatures more exact than 9°F (5°C) ranges, the maximum scope for activity seems to lie between 77 and 86°F (25 and 30°C). The direct result, also seen in Figure 6, is that the maximum swimming speeds of both fingerling and adult largemouth bass decline at temperatures above about 86°F (30°C). A similar pattern of swimming speed capabilities over a wide range of temperatures was shown for smallmouth by Larimore and Leever (1968). Differences between largemouth and smallmouth bass do not show up from the wide temperature intervals examined.

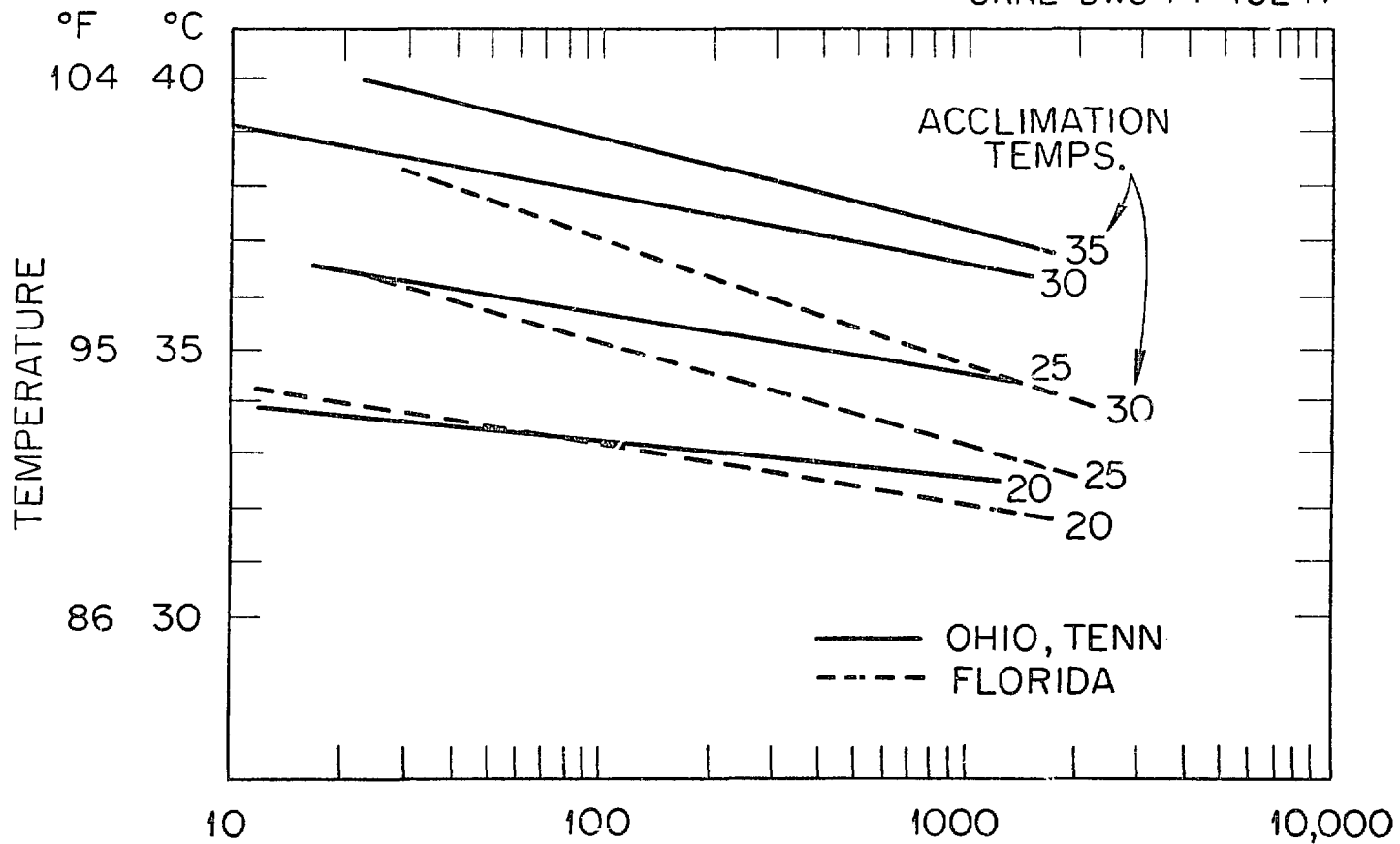
Tolerances of Extreme Temperatures

Survival of largemouth bass at extreme high temperatures was found by Hart (1952) to depend upon three main factors (as is typical of all fish that have been studied): (1) the initial acclimation or holding temperature, (2) the exact test temperature, and (3) the duration of exposure to the test temperature (Fig. 7). In addition, Hart discovered an odd relationship between the Florida subspecies and its northern cousin - that the southern fish are more sensitive than the northern ones, particularly at higher acclimation temperatures. The highest temperature which allowed long-term survival of the northern fish was about 97.7°F (36.5°C); for the southern fish it was about 93.2°F (34°C). These studies should be repeated to determine their accuracy.

Largemouth are also sensitive to temperature decreases. Cold kills are rare in natural environments (where northern bass survive to the

Fig. 7. Time to 50% mortality of adult largemouth bass at high temperatures when transferred from several acclimation (holding) temperatures. Fish from Ohio and Tennessee (solid line) are compared with fish from Florida (dashed line). From Hart 1952.

ORNL-DWG 74-13247



freezing level), but large, abrupt temperature decreases that occur when power plant thermal effluents are suddenly cooled (sudden plant shutdown) may cause indirect deaths of young bass (see Coutant et al. 1974 for field reports). Our own research has shown that young bass become more susceptible to predation by larger fishes when they experience temperature decreases of more than about 9°F (5°C) (Coutant et al. 1974).

Detailed studies of heat tolerance of smallmouth have apparently not been done (although Larimore and Duever 1968 noted an unpublished upper lethal temperature of 38°C or 100.4°F). Studies of tolerance to rapid cooling by Horning and Pearson (1973) revealed that decreases amounting to 21.6-25.2 degrees (F) (or 12-14°C) could be tolerated with 90-100% survival when the young fish were acclimated to a wide range of initial warm temperatures.

Temperature Selection

After a decade of pollution control research emphasizing directly lethal effects of pollutants, we are realizing that fish actively select or avoid certain conditions, including temperature (Coutant, in press). Such behavior may determine the conditions which fish actually experience far more than the physical and chemical conditions recorded at fixed monitoring stations.

Largemouth

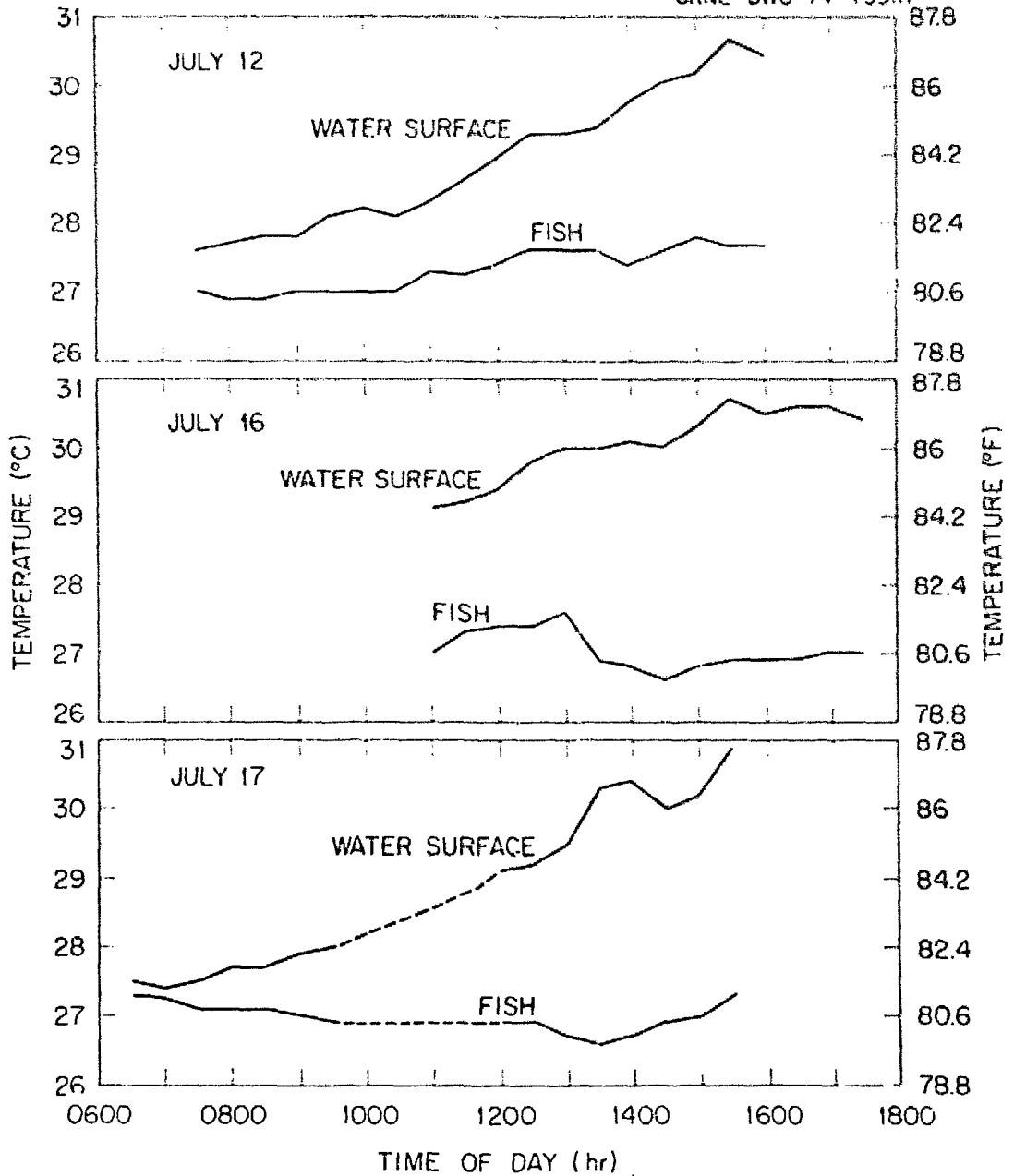
In cold climates, largemouth seem to overwinter in deep water which is then the warmest available (Webster 1954). My own research using temperature sensing ultrasonic tags in East Tennessee shows that

as the spring sun begins to warm the surface layers, bass move into the shallows, and are generally found in nearly the warmest water available (Coutant, in press). Here they spawn, and the adults generally remain until the surface waters reach near 80°F (27°C). When seasonal or daily temperatures at the surface rise above about 80°F, fish that have cooler water below them in stratified lakes swim deeper, following the water that stays near 80°F as long as oxygen is available. Dendy (1948) saw this selection in his gill netting studies of Norris Reservoir, Tenn. in the 1940's. Fish with only shallow water near them will move laterally out of shallow areas into the deeper spots where the 80°F temperatures can be found (Fig. 8). Often the daily changes in surface temperature of midsummer oscillate between the preferred 80°F in early morning and much higher temperatures in late afternoon causing the fish to appear in the surface fishing spots in the morning and disappear from the average floating-plug fisherman's domain in the afternoon. In one cooling reservoir in Texas, largemouth bass sought out cooler shoreline zones in summer mornings when the remainder of the reservoir had temperatures exceeding 100°F (37.8°C) (Smith 1972). As autumn temperatures decline below the preferred range near 80°F, bass again reappear in surface waters that are warmest. Superimposed on this general pattern are short-term excursions into water both above and below the preferred temperature.

Small bass seem to prefer warmer water than do adults. Casual observations by many people in the field indicate that young bass remain in the shallows in summer long after the adults have left.

Fig. 8. Temperatures selected by an adult largemouth bass (fish) and water surface temperatures at half-hour intervals for three days in July 1974, K-25 pond, Oak Ridge, Tennessee, USA (from Coutant, in press).

ORNL-DWG 74-7551R



Neill (1971) found bass 2.8-3.9 in. (72-99 mm) long selecting 84.7-87.6°F (29.3-30.9°C) in the field and about 84.2°F (29°C) in the lab. The oft-quoted temperature preference of 86-89.6°F (30-32°C) that originated in unpublished research at the University of Toronto (Ferguson 1958) was determined for young fish. Laboratory determinations of preferred temperatures in gradient tanks are generally conducted on fingerlings as a matter of convenience. Several comparisons of young and adults of other species also indicate that young fish tend to select somewhat warmer water than do adults (see discussion in Coutant, in press).

Winter attraction of largemouth bass to heated power plant discharges seems fairly common. One of the most studied sites is at an AEC production reactor cooling pond in South Carolina (Bennett 1971, Gibbons et al. 1972, Clugston 1974). Here, adult largemouth bass congregate in very large numbers in the heated discharge arm. The same discharge does not attract such large numbers of bass when the reactor is not operating and the effluent releases cool water. So many bass congregate there that the condition of some is markedly reduced, apparently due to depletion of available food. Other bass show high mobility, and readily move between the warm arm of the reservoir and cooler areas. There was no clear attraction to 80°F (27°C) when this was available at the immediate discharge point, suggesting that the particular aspects of currents, shelter, etc. influence the small-scale distribution more than does exact temperature. In summer, body temperatures of largemouth were in the 80.6-86°F (27-30°C) range.

A day-night difference in largemouth distribution near a Wisconsin power plant outfall was seen by Neill (1971). Fish gathered near the

outfall in the daytime (84.7 to 89.6°F or 29.3 to 32°C) and spread out more evenly between heated and non-heated areas at night (selection in all areas was 79.7 to 84.4°F or 26.5 to 29.1°C). Residence times overall in that thermal plume were short - 1 week or less. Some largemouth bass there had body temperatures exceeding 86°F (30°C).

Reflecting on the temperature responses seen earlier for feeding rate, growth, metabolism and swimming speed of largemouth bass (Table 1), we see that they all are maximum near 77-86°F (25-30°C), and that this is the general range selected by fish in the field. Such similarity could hardly be coincidence. Largemouth would seem, then, to actively select, if available, those temperatures at which it performs best. One of the most interesting directions for further research would be to define the levels of other environmental requirements - cover, oxygen, silt, current, food availability, and so forth - that override this apparent general rule.

Smallmouth

Field observations in the summer suggest that smallmouth bass prefer temperatures near 70°F (21°C) in both lakes (Hile and Juday 1941) and streams (Hallam 1958). In Ontario waters, this has meant that the fish inhabit the shorelines in summer (Turner and MacCrimmon 1970). In warmer Tennessee, however, smallmouth seek deeper water (Stroud 1948). An occasional smallmouth has been caught in water as warm as 92.3°F (33.5°C) (Gammon 1973) and 98°F (36.7°C) (Churchill and Wojtalik 1969), but these must be exceptions not generally tolerated for long.

As with largemouth, there seems to be conflicting laboratory data. Fry (cited by Ferguson 1958) determined a preference by juveniles for 82.4°F (28°C). This, too, may reflect a real difference due to age. Barans and Tubb (1973) found underyearlings in a horizontal gradient to select 84.2-87.8°F (29-31°C) in summer, 78.8-86°F (26-30°C) in fall, 75.2-82.4°F (24-28°C) in winter and 71.6-82.4°F (22-28°C) in spring--all higher than the field records would indicate. Adults, too were higher in the laboratory: 86-87.8°F (30-31°C) in summer, with somewhat lower preferences in other seasons. Peak (1965) found his adult smallmouth selecting 87.8°F (31°C) in a horizontal laboratory gradient. The conflicting data may not be resolved without more extensive field studies.

Watt's (1959) observation that smallmouth congregate in certain (unspecified) locations in South Bay, Ontario, in winter and Webster's (1954) statement that in Cayuga L., N.Y., they lie in fissures in the bottom may indicate a winter selection for warm spots caused by springs.

Spotted

Spotted bass seem to prefer temperatures near 75.2°F (24°C) in midsummer according to studies by Dendy (1948) and Stroud (1948) on Norris Reservoir, Tenn. Stroud indicated that the spotted bass go to deep water in late spring and are not caught by the surface plug fishery in summer. The range 72.5 to 80.6°F (22.5 to 27°C) was inhabited by spotted bass in the Wabash River, Indiana studied by Gammon (1973). In a section of the White River, Indiana, Benda and Proffitt (1974) showed a reduction by 1/2 in the numbers of spotted bass inhabiting the zone downstream of a thermal effluent compared to upstream areas throughout the years 1969 and 1970, a change they attributed to temperature differences.

Temperature and Fishing Success

A few studies have attempted to relate temperatures to fishing success. Mostly, however, a fisherman's folklore has developed about temperature, some of which may be very valid and can be related to (and help support) the temperature relationships that have been studied in detail. A good compilation of this folklore is needed. This is an area where scientists can learn a great deal from sportsman.

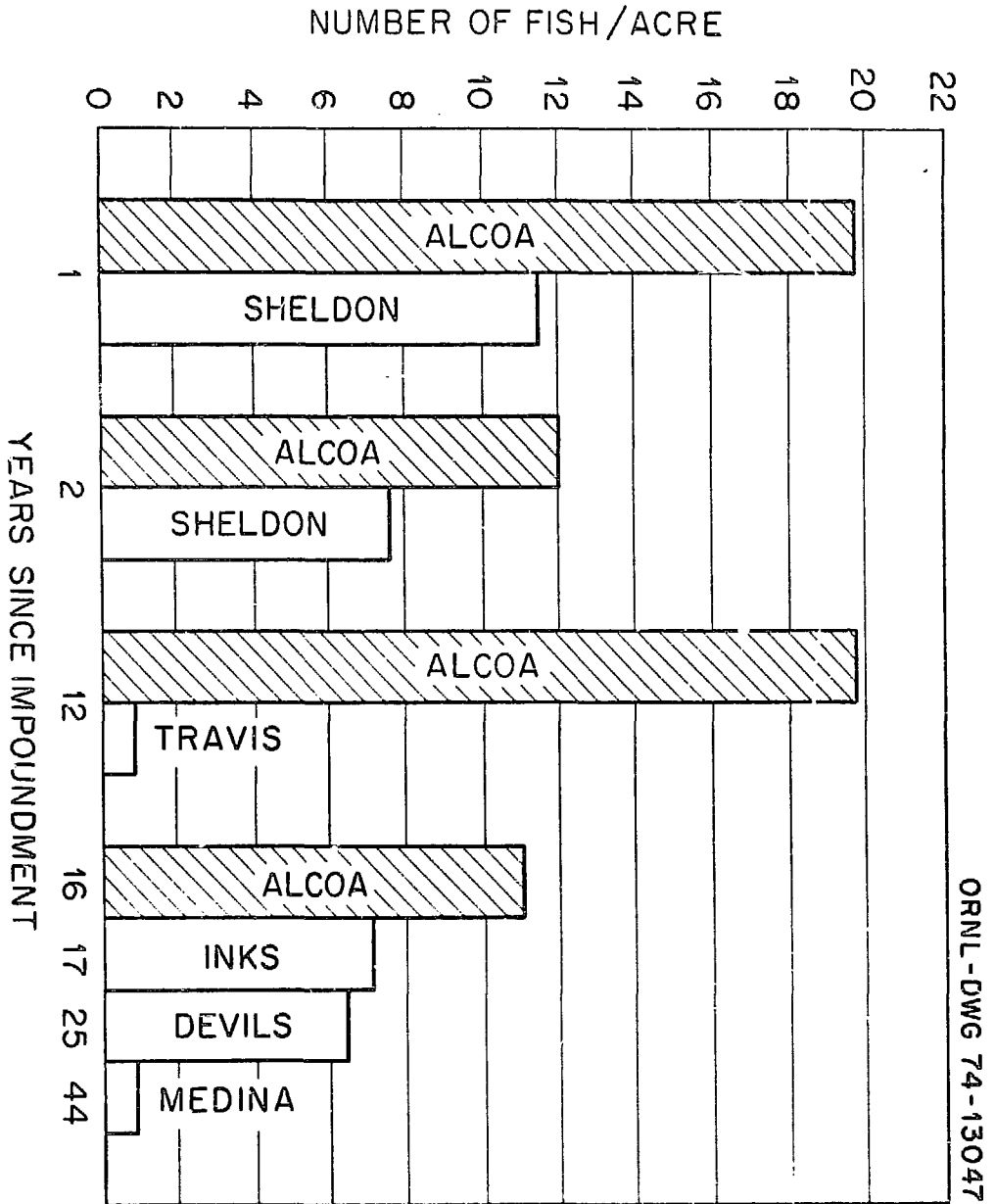
Largemouth

In a systematic casting study in midday in a South Carolina cooling reservoir, Gibbons et al. (1972) caught significantly more bass per cast in the vicinity of a heated effluent (average maximum temperature 74.8°F, or 23.8°C) than when the effluent was unheated or at an unheated reference location on the same reservoir. This may have been due in part to extreme crowding and undernourishment of some fish in the warm area in a lake that is generally closed to fishing.

Largemouth bass catches in some Texas cooling lakes have been greater than in reservoirs not used for power plant cooling (Fig. 9). There were also better game/trash fish ratios in the warmed lakes. Detailed comparisons of populations (sizes, stocking rates, etc.) and water quality parameters should be conducted and made generally available.

Many reports have noted declines in catchability of largemouth at temperatures greater than about 80°F (27°C). Part of this trend may be due to decreased feeding by the bass at these temperatures as suggested earlier, but part may also be due to fish leaving the traditional shoreline fishing areas and seeking cooler water. Reports of great success by fishermen who have found a "deep hole" in midsummer are also common.

Fig. 9. Number of largemouth bass per acre taken by anglers from six reservoirs in Texas, one of which (Alcoa) was used for power plant cooling. Adapted from Drew and Tilton 1970 and based on Texas Parks and Wildlife Department creel census reports and fish catch records.



Smallmouth

Cool-weather smallmouth fishing seems to be improved by sources of warm water. Catch-per-unit-effort for smallmouth was shown in one study in South Bay, Lake Huron (Watt 1959) to rise markedly in temperatures below about 55°F (12.8°F). This was thought by Watt to be a result of smallmouth bass tending to congregate in certain unspecified locations in cold weather (warm springs?) and of fishermen who knew by experience where the aggregations occurred making larger catches. More thorough study of this phenomenon seems desirable. Trembley (1960) reported increased fall, winter, and spring fishing success for smallmouth in the Delaware River near a thermal discharge.

The annual catch in a creel survey of selected fishermen on the Potomac River near a power plant discharge showed more smallmouth caught in the heated waters (Elser 1965). The annual catch ratio among one heated and two unheated stretches of river was 49:18:34. Heated and non-heated areas were carefully given equal fishing coverage. Smallmouth fishing did decline in summer, and no fish were caught during the warmest period. It is interesting that the annual heated area catch was larger in spite of the negligible catches there in summer. This relationship should probably be balanced by overall fisherman demand in the area (which the study did not evaluate) which may have been greater in the poor summer months.

Population Success

As in the adage that the proof of the pudding is in the eating, the value of all of the foregoing research (Summarized in Table 1)

Table 1. Summary Table for Important Temperatures for Largemouth and Smallmouth Bass.

		Spawning	Incubation ¹	Feeding (Maximum)	Growth		Metabolic Scope (Maximum)	Swimming Speed (Maximum)	Incipient Lethal Temp.	Preferred Temp.	
					Range	Optimum				Adults	Young
Largemouth	°F	66-68	59-77	68-80.6	50-95.9	80.6	77-86	77-86	97.7	80.6	86-89.6
	°C	18.9-20	15-25	20-37	10.35.5	27	25-30	25-30	36.5	27	30-32
Smallmouth	°F	62	55-79	?	50-?	?	?	77-86	?	70	84.2-87.8
	°C	16.7	12.8-26.1	?	10-?	?	?	25-30	?	21	29-31

¹ Males may desert nests when temperature falls below spawning temperature, causing death of eggs.

rests in evaluating and predicting the overall success of bass populations in the wild. Here temperature must be placed in perspective with all other important habitat requirements, and the many temperature effects throughout the life cycle must be evaluated for relative importance. Only largemouth and smallmouth bass populations have been examined in any detail.

Largemouth

The distribution pattern of native and introduced populations of largemouth throughout the world analyzed by Robbins and MacCrimmon (1974) indicates a wide range of suitable temperature conditions, but nonetheless a clear importance of water temperature for success of the species. The present distribution ranges from the cold regions of southern Canada to the equatorial waters of Ecuador, Kenya and Uganda, and altitudes ranging from sea level to 7000 ft (2100 m). The thermal tolerance range does not seem to be exceeded even in the tropics, where other factors such as reproductive failure limit populations. Only three other factors: turbidity, predation by native species (where introductions have failed), and avoidance of fast-flowing waters appear to be more important than temperature in determining population success.

Wherever normal spawning temperatures are provided, largemouth seem to flourish, even when the relatively wide annual fluctuations in temperature of its native temperate range are not present. It had been thought that a clear annual cycle of temperature was needed by largemouth in which a winter dormant period was followed by conditions favorable for spawning and active growth. Successful introduction to the more uniform, warm climates of Africa, the Mediterranean and

The importance of suitable spawning temperatures for success of largemouth has been emphasized by Robbins and MacCrimmon (1974) based upon the relationship between population success and the temperature gradients with increasing altitude in both the Old World and the New. In Kenya, favorable temperatures occur between 4000 and 7000 ft (1200 and 2100 m) above sea level. In nearby Uganda, the greatest success is above 6000 ft (1800 m). At increasingly northern latitudes successful spawning occurs at correspondingly lower altitudes: in Morocco from sea level to 5000 ft (1500 m); in France only below 2000 ft (600 m). Lack of waters warm enough for spawning apparently prevents extension of the range in Europe north of the Alps. Similarly, in central Mexico largemouth are restricted to waters above 5000 ft (1200 m) while in Colorado the species is limited to valleys at altitudes below 9000 ft (2700 m).

Research in Canada discussed by Robbins and MacCrimmon (1974) also suggests that low summer temperatures may be effective in restricting establishment of largemouth. Even where spawning temperatures are reached in the northern limit of natural dispersal, the largest populations of largemouth are found in the shallow bays or smaller lakes that provide warmer summer temperatures than larger, open waters. No clear cut-off temperature is apparent, however.

Smallmouth

Robbins and MacCrimmon's (1974) examination of ecological factors associated with results of smallmouth introductions to waters throughout the world revealed water temperature to be the most important single factor limiting distribution. As with largemouth, the necessary

spawning temperatures must be reached before the species can become self-sustaining in northern waters. When temperatures do not rise much above the spawning level, excessive predation on the slowly growing fry seems to restrict populations. Smallmouth populations are also restricted by warm temperatures (e.g., in downstream sectors of rivers and in warm-water reservoirs) but the reasons are not clear. Some smallmouth populations seem to thrive in Southeast Asia where temperatures may not drop seasonally to the typical spawning temperature. Excessive turbidity appears to be the other predominant environmental restriction.

Implications for Management of Heated Waters

Much of the recent concern over temperature requirements of the basses stems from questions about the management of thermal discharges from the electricity generating industry. So it is logical to attempt some conclusions and speculations regarding this management. Past experience should be our guide in establishing rational thermal limits to protect the bass and in proposing future management of both thermal discharges and fish populations.

First, there has been exceptional fishing success for both largemouth and smallmouth in areas of heated discharges in the cooler months. This has been popular with sports fishermen, as anyone who has fished near a thermal outfall in winter can attest. One of the biggest problems in sports fishery management is overcome - getting the fishermen and the fish together in the same spot. This cool weather advantage usually disappears in summer when discharge temperatures are too high and

the fish leave. Nonetheless, the yearly balance seems to be in favor of better bass fishing near thermal discharges. This we ought to keep, if possible.

Second, largemouth bass populations seem to do better in warmer waters, suggesting that managed, warm-water lakes should be a benefit to largemouth fishing. The hazard of rapid chilling during spawning is minimized when warmed water is available, as hatchery pond experience has shown. Growth rates are fastest at relatively high temperatures (near 80°F) and growth in most bass populations seems to be limited by a short growing season. Thermal additions into well managed cooling reservoirs should allow exceptionally rapid growth compared to nearby lakes with natural temperatures. Since cooling lakes are far cheaper for much of the utility industry than cooling towers, the managed cooling lake could serve both for efficient power plant cooling and exceptional largemouth bass fishing. This is not a popular view lately, but it seems well supported by the research data.

Smallmouth bass, with their apparent lower temperature requirements (although data are skimpier), probably would not benefit by thermal additions except in the northern parts of the range. There the data show better year-class strength and better catches in warm climatic cycles than in cool ones. Greatest care should be taken with any warming of the waters in the southern part of the range where smallmouth are desired. The species does not do well in most reservoir environments, however, so these areas probably should be managed for largemouth fisheries instead.

Third, information on growth, metabolism and swimming ability of largemouth suggest that temperatures above about 86°F (30°C) are too

high for healthy fish. Wherever the bass cannot follow their tendency to avoid these higher levels, then thermal standards should be established to limit long-term temperatures (24 hr averages) to 86°F. Short-term high temperatures (e.g. peak temperatures of late afternoon) can safely exceed 95°F (35°C) without apparent damage. The presence of cool, well-oxygenated refuge areas may be the most important factor determining success of bass populations in warm cooling lakes in midsummer. New cooling reservoirs ought to be designed to provide this feature. Well-designed cooling reservoirs could actually provide more favorable midsummer habitats for bass than many natural lakes with hot surface waters and oxygen-poor deeper layers.

While information for smallmouth is less complete, the upper limit for healthy populations of this species would certainly be no higher than that for largemouth. Further research may show that it should be somewhat lower. Spotted bass seems to be intermediate between large and smallmouth in thermal requirements wherever it has been examined.

Fourth, the potential hazard of stressing fish with cold shock in power plant discharges that cool quickly should be minimized. Engineering provisions can and should be made to (1) limit the area of maximally heated water and thus the number of fish that can become warm-acclimated, (2) ensure slow cool-down even though the power plant shuts down rapidly, or (3) increase the reliability of constantly-warmed water by locating more than one power plant unit (within limits) at the site, for it would be rare for more than one unit to shut down at the same time. The industry has been preparing guidelines to accomplish these objectives.

In summary, the temperature relationships of largemouth bass suggest that this species is unlikely to be damaged by thermal discharges designed with the needs of the bass in mind. Instead, the greater growth at more southern latitudes may be emulated in well managed cooling lakes to enable both highly productive largemouth fisheries and efficient power plant cooling to go hand in hand. Smallmouth fishing may be similarly improved in managed situations near the northern limits of its range (e.g., southern Canada, northern U.S.), but special care should be taken in the South. There are too little data available on the other centrarchid basses to even hazard a guess.

The key to proper design of thermal discharges and effective management of bass populations and their sport fisheries in cooling reservoirs is full understanding of the fishes thermal requirements, and behavior. This review shows that we still have a lot to learn, even about the most-studied of the basses, the largemouth and smallmouth.